Missing Data

Learn to understand, design, build, test and explain AI systems responsibly.

Project: Transparent Book Recommender

Recall and choose your starting point

> Record your starting explanation for Missing Data before opening notes. Show any later correction separately.

Try the recall before opening notes. Then use a different colour or a labelled second line to correct your explanation.

Noise and Outliers

Without opening the old lesson, explain one key idea from Noise and Outliers. Give an example and a mistake that the idea helps you avoid.

> **My remembered idea, example and correction**

Choose Informative Questions

Without opening the old lesson, explain one key idea from Choose Informative Questions. Give an example and a mistake that the idea helps you avoid.

My remembered idea, example and correction

Accuracy as a Fraction

Without opening the old lesson, explain one key idea from Accuracy as a Fraction. Give an example and a mistake that the idea helps you avoid.

My remembered idea, example and correction

WORKBOOK · 2 OF 21

Practise with fading help

> Complete the supported Missing Data trace and record the first step you can explain without help.

Use W2: Trace filtering for **Missing value**s and **validation**. Record the answer once; the lesson and workbook refer to the same attempt.

Given inputs	Procedure I chose	Middle steps	Result and check

The step I completed without help, and the hint I needed

WORKBOOK · 3 OF 21

Individual reasoning check

> Check the proposed Missing Data explanation against the supplied facts before discussing it.

Use the worked case for **Missing value**s and **validation**. Proposed explanation: “A successful run proves the conclusion”. Decide whether this explanation is supported.

Point to the step or supplied fact that decides; explain your repair if needed.

My explanation before partner discussion

WORKBOOK · 4 OF 21

Independent transfer case

> Write one first attempt at the reserved independent case before feedback. Repeated copies of that case refer to the same recorded attempt.

Independent case: Validate independently

Write a filter for count values that must be integers from 0 to 10. **Process** {count: 3}, {}, {count: 0}, {count: 11}, {count: "3"}, using quoted keys in Python. Report accepted values and rejected count.

> **My independent explanation, trace and limits**

Attempt this case before feedback. Record any help. Earlier examples below are further practice; a case already practised with feedback cannot establish fresh transfer.

WORKBOOK · 5 OF 21

Your learning target

> Connect the Missing Data target to a trace and a justified conclusion.

Scores award 2 for preferred genre and 1 for short length. A book matches genre but page count is missing. Compare rejecting it, assigning zero length points and asking for pages.

WORKBOOK · 6 OF 21

Solve the given case

> Solve the supplied Missing Data case and show the deciding evidence.

Genre is known, but page count is missing and length affects recommendation score.

Choose transparent handling.

Show your trace, calculation or **evidence**

Write or draw your reasoning and evidence.

WORKBOOK · 7 OF 21

Check a changed case

> Apply the procedure to the changed case and record why the conclusion follows.

A book's page count is blank. Is blank the same as zero? Give an honest way for a short-book recommender to respond.

My independent answer and reason

Write or draw your reasoning and evidence.

Week 18 Test and reflect

WORKBOOK · 8 OF 21

Design a revealing test

> Specify a changed input, expected result and observation separately.

Create a changed **input**, boundary case or missing-information case. Write the expected answer or safe response before testing.

Test record	My evidence
Changed input or condition	
Expected answer or fallback	
Observed result	
Why the result occurred	

Project checkpoint

> Justify what Missing Data adds to Transparent Book Recommender and keep the evidence.

Use ‘no preference’ only when the reader confirms it; otherwise ask or explain uncertainty.

The project change and **evidence** I will keep

> **Write or draw your reasoning and evidence.**

WORKBOOK · 10 OF 21

Self check

> Judge explanation, evidence, transfer and limits separately; attach the work behind your judgement.

- I can explain the mechanism.
- I can show evidence.
- I can handle a changed case.
- I can name a limit.

Mark each as not yet, with help or independently, and attach supporting work.

Teacher review _____ Next step _____

WORKBOOK · 11 OF 21

Transfer practice from the original programme

> Use this earlier case as transfer practice. A case already discussed with feedback is not fresh assessment evidence.

Scores award 2 for preferred genre and 1 for short length. A book matches genre but page count is missing. Compare rejecting it, assigning zero length points and asking for pages.

My independently reasoned answer

Record support used: none, hint or teacher explanation.

WORKBOOK · 12 OF 21

W1 Distinguish zero and unknown

> Distinguish zero and unknown. Apply the supplied Missing Data facts and keep your reasoning.

Explain the difference between {"minutes": 0}, {} and {"minutes": None}.

My reasoning and evidence

WORKBOOK · 13 OF 21

W2 Trace filtering

> Trace filtering. Apply the supplied Missing Data facts and keep your reasoning.

Predict accepted values and **rejection**s for minutes 4, 0, 61 and "4".

My reasoning and evidence

WORKBOOK · 14 OF 21

W3 Repair a condition

> Repair a condition. Apply the supplied Missing Data facts and keep your reasoning.

A program uses if value to accept durations. Give a counterexample and the correct rule.

My reasoning and evidence

WORKBOOK · 15 OF 21

W4 Validate independently

> Attempt before feedback. If this repeats the selected independent case, reuse that first attempt; otherwise keep a separate record. Validate independently. Apply the supplied Missing Data facts and keep your reasoning.

Write a filter for count values that must be integers from 0 to 10. **Process** {count: 3}, {}, {count: 0}, {count: 11}, {count: "3"}, using quoted keys in Python. Report accepted values and rejected count.

My reasoning and evidence

W5 Report the limitation

> Report the limitation. Apply the supplied Missing Data facts and keep your reasoning.

> Only 2 of 5 supplied durations were accepted. Why should the report include this?

My reasoning and evidence

Feedback, return check and learning record

> Record the first repaired step and a different case to revisit after a delay.

Evidence	My record
Worked alone / hint / model	
First step I repaired and why	
New case checked later; date and result	
What I can now explain without notes	
One remaining question	

Revisit this idea with your teacher after a delay. Familiarity is different from explaining and using it on a changed case.

WORKBOOK · 18 OF 21

Pictorial case practice: Missing Data

> A fictional catalogue records counts of available copies under a strict input contract. Explain the labelled step before checking the changed condition.

A fictional catalogue records counts of available copies under a strict **input** contract.

Accept count only if type(value) is int and 0<=value<=8. Missing fields are invalid, and Booleans are not counts.

Rows have count 0, count 4, absent count, count "4", count True and count 9.

AI ACADEMY / AI-2/18

Missing Data

Start with the supplied facts

Follow each arrow. Ask which fact allows the next step.

1. Retrieve carefully

Use row.get("count"); absent field gives None, distinct from 0.

2. Check exact type

Integers 0,4 pass; text "4", Boolean True and None fail.

3. Check range

Integer 9 fails maximum 8.

4. Report validation

Accepted counts [0,4]; four rows rejected with recorded reasons.

CHANGE ONE THING

Replace text "4" with integer 4 through an authorised conversion.

Explain the changed result using the matching lesson.

Follow the labelled picture with the matching explanation. Open the vector diagram

Why is the Boolean True rejected even though arithmetic can treat it like 1 in Python?

My picture-based explanation and changed-case prediction

Record the help used

Support used: alone / hint / worked example. This record is practice **evidence**.

Project connection: Validate recommender stock and numeric features explicitly, recording zero stock as measured zero rather than missing.

Project portfolio milestone

> Keep this week's input, deciding step, result and limit together in the Transparent Book Recommender record.

Project: Transparent Book Recommender

Current milestone: Test changed cases. Keep normal, boundary and failure cases; compare expected and observed results.

Validate recommender stock and numeric features explicitly, recording zero stock as measured zero rather than missing.

Evidence to keep	My record
This week's input and deciding step	
Prediction and actual result (or labelled paper trace)	

Evidence to keep	My record
One change since the previous checkpoint and why	
One error or limit and the next test	

> **Explain what this week adds to your project. Reuse last week's record rather than starting again.**

WORKBOOK · 20 OF 21

Supported practice and learning evidence

> Try the supported questions before comparing with hints and feedback. Record what you repaired and the evidence you retained.

Use this supported practice once, in the browser or on paper. Keep the reserved independent assessment for a separate first attempt before teacher feedback.

A fictional catalogue records counts of available copies under a strict **input** contract.

Practice 1: explain the deciding evidence

Which labelled step matches this detail? Accepted counts [0,4]; four rows rejected with recorded reasons.

1. Report **validation**

2. Retrieve carefully

3. Check range

4. Check exact type

My selected conclusion, deciding fact and explanation

Hint · try first

Hint if needed: Read the steps in the pictorial case. Match the supplied detail to the stage that performs or records it.

Feedback · compare after trying

After your attempt, compare with the supported explanation: Report **validation** Accepted counts [0,4]; four rows rejected with recorded reasons. Testing if value would wrongly discard zero, which is a real valid count.

Python’s isinstance(True,int) is true, so the stated exact-integer contract uses type(value) is int to reject Boolean **input**.

Keep missing, type-invalid and range-invalid reasons separate rather than turning every rejected value into zero.

Practice 2: explain the deciding evidence

Changed case · predict first

Changed-case practice: Replace text "4" with integer 4 through an authorised conversion. Which conclusion is supported by the supplied facts?

1. We can decide the result without examining the changed **input** or the procedure.

2. This one example guarantees correct results for every future **input**.

3. Accepted values become [0,4,4]; three rows remain rejected.

My selected conclusion, deciding fact and explanation

Hint · try first

Hint if needed: Use the changed condition and the deciding step in the pictorial trace. The answer must say what follows from those facts.

Feedback · compare after trying

After your attempt, compare with the supported explanation: Accepted values become [0,4,4]; three rows remain rejected. The converted value now satisfies both type and range checks.

Repair a missing building block

Review Noise and Outliers (ai-2/17). Goal: Calculate mean and median. Short practice: If 18 was genuine, which summary better describes the middle observed task, and what extra fact should be shown?.

Return to this week and explain what you changed.

Review Choose Informative Questions (ai-2/15). Goal: Fit a root question and leaf **prediction**s from **training** examples. Short practice: Choose the best original question and explain why its more balanced split matters here..

Return to this week and explain what you changed.

Review Accuracy as a Fraction (ai-2/12). Goal: Explain why overall accuracy can hide failure on a less common class. Short practice: Why is 100% repair detection not sufficient to declare the changed classifier best?.

Return to this week and explain what you changed.

Predict, run and debug

The browser lab lets you edit the supported Python examples already included in this lesson. Predict first, run, inspect the actual **output** or error, change one **input**, then run again.

Keep code, **prediction**s, actual results and a reasoned revision together. On paper, label your result as a trace rather than an execution.

Keep your completed work

Use Download completed work in the hosted lesson to export your answers, reflections, practice feedback and code-run **evidence**. Open the HTML copy to read or print it.

Drafts stay in this browser when storage is available; clear drafts on a shared device.

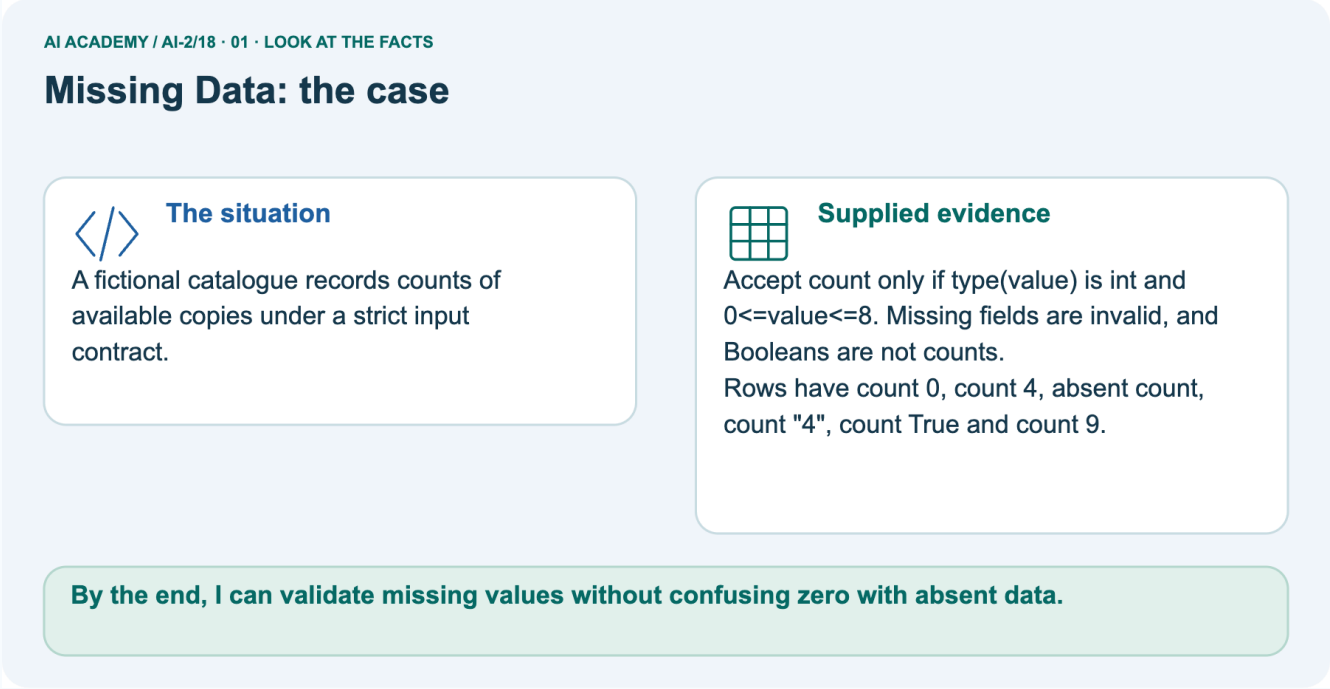
What I repaired, the evidence I kept and the next test

WORKBOOK · 21 OF 21

Visual concept journey

> Explain each Missing Data picture from its labels, then record a relevant limit.

01. Missing Data: the case



Follow the labelled picture with the matching explanation. Open the vector diagram

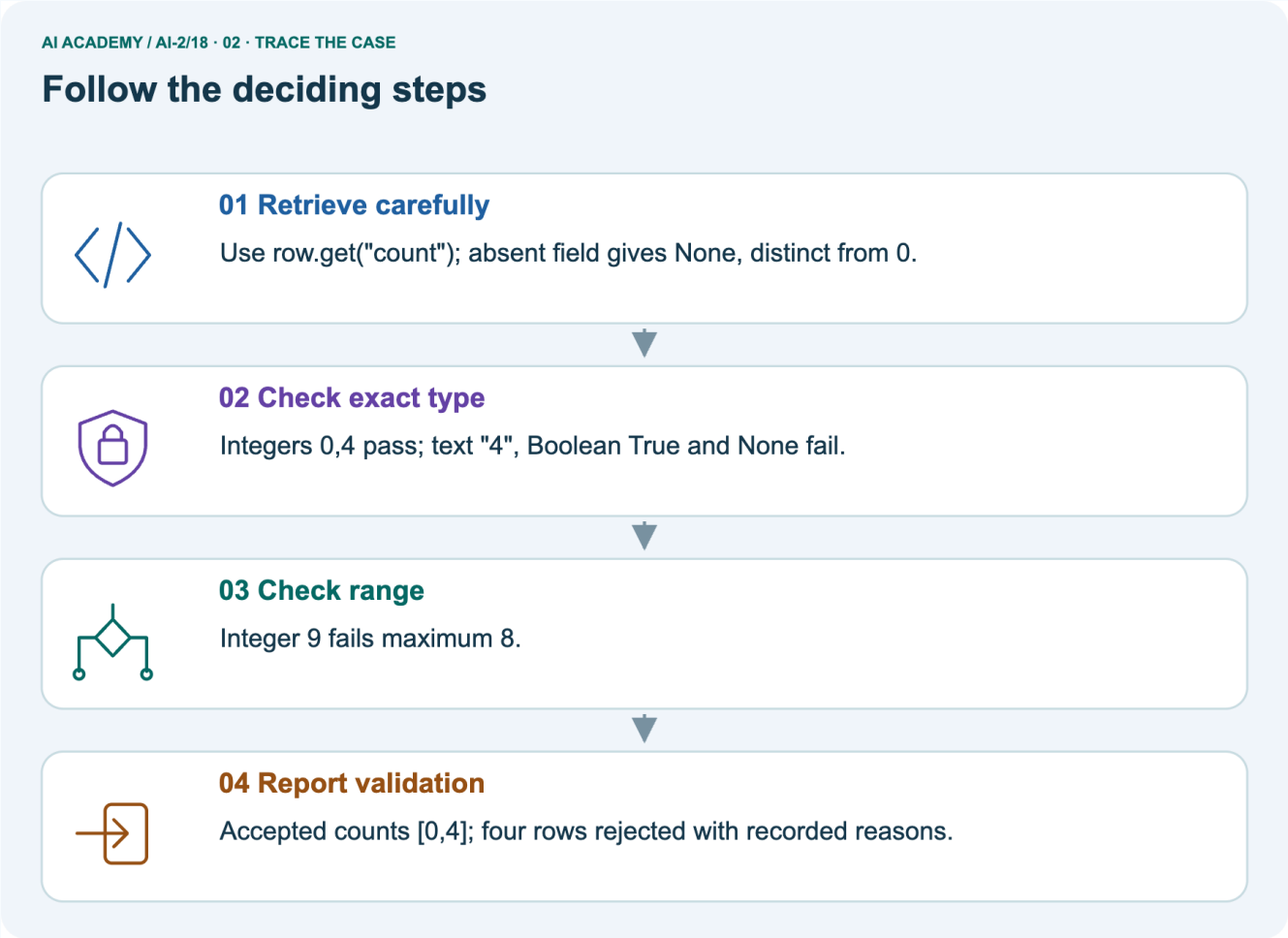
A fictional catalogue records counts of available copies under a strict **input** contract. Supplied facts: Accept count only if type(value) is int and 0<=value<=8. Missing fields are invalid, and Booleans are not counts.

Rows have count 0, count 4, absent count, count "4", count True and count 9.

> Point to the given fact that matters most. Explain why it matters.

My picture explanation, deciding evidence and one limit

02. Follow the deciding steps



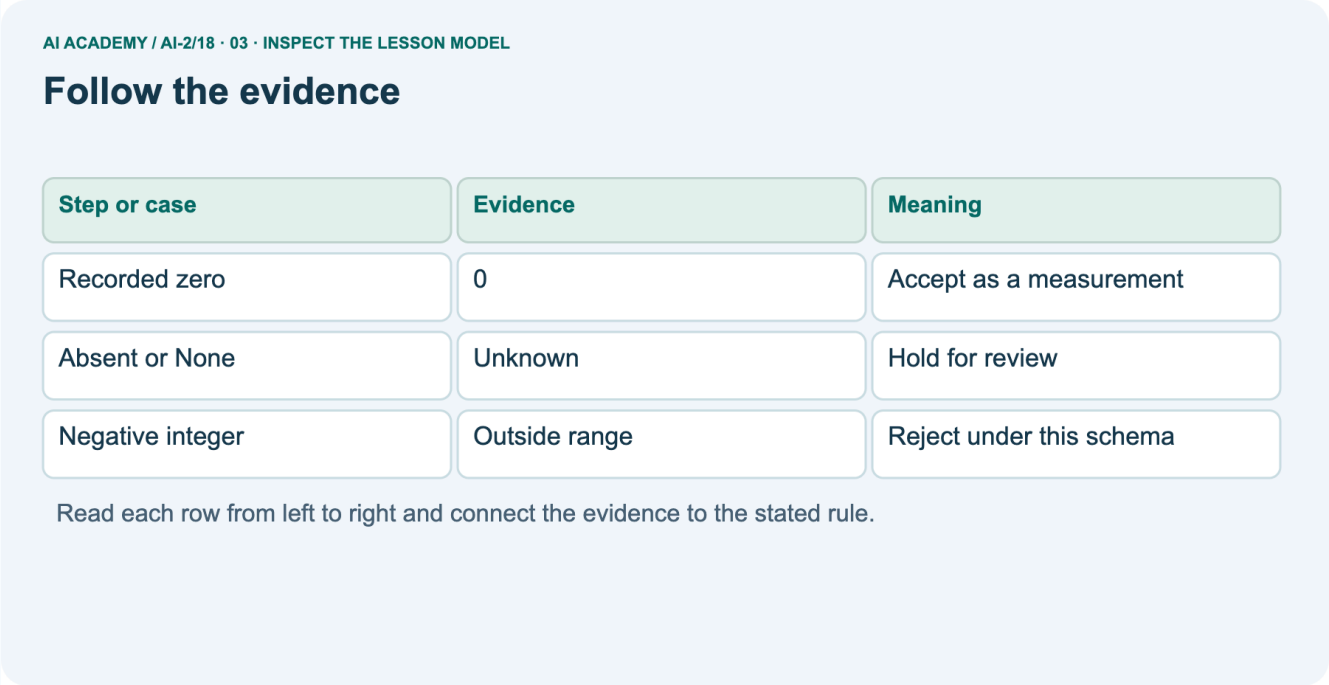
Follow the labelled picture with the matching explanation. Open the vector diagram

Retrieve carefully: Use row.get("count"); absent field gives None, distinct from 0. → Check exact type: Integers 0,4 pass; text "4", Boolean True and None fail. → Check range: Integer 9 fails maximum 8. → Report **validation**: Accepted counts [0,4]; four rows rejected with recorded reasons.

> Point to each numbered stage and explain which recorded **input** it uses.

My picture explanation, deciding evidence and one limit

03. Follow the evidence



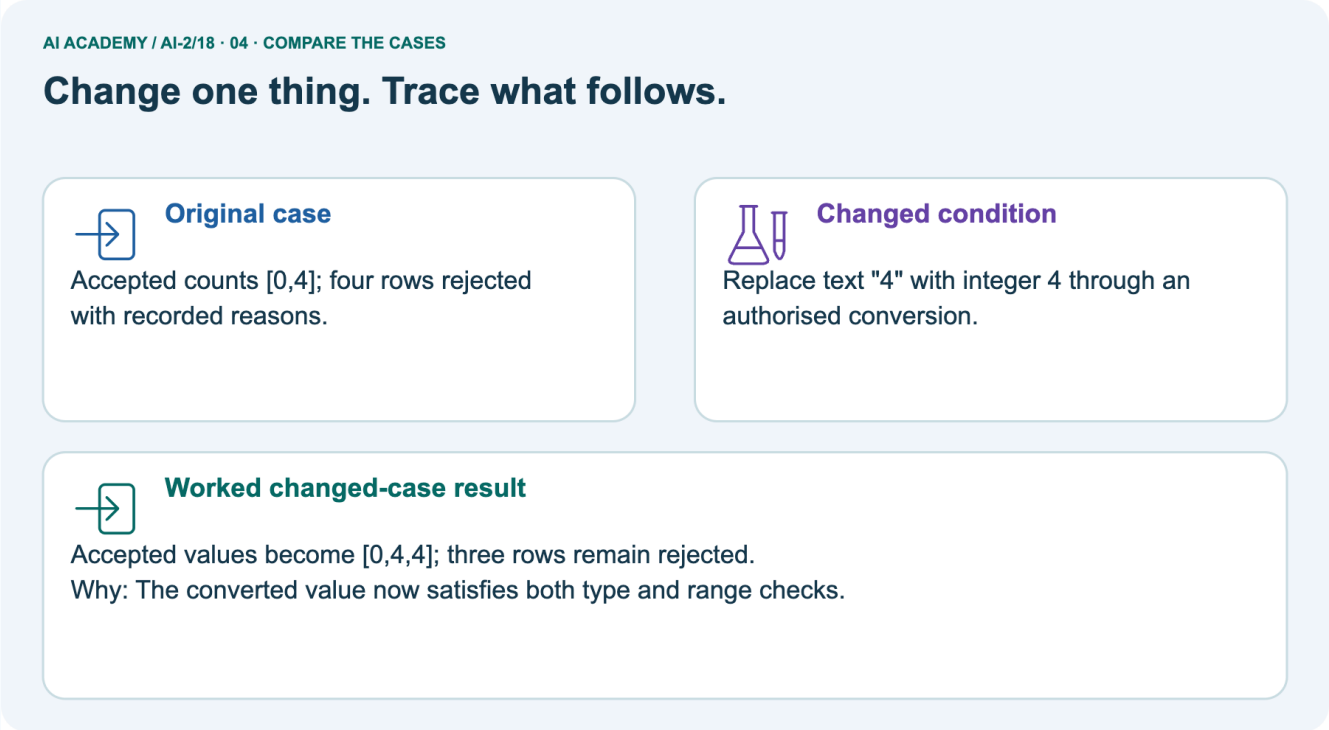
Follow the labelled picture with the matching explanation. Open the vector diagram

Follow the **evidence**. Recorded zero | 0 | Accept as a measurement. Absent or None | Unknown | Hold for review. Negative integer | Outside range | Reject under this schema

Read the labels and values. Explain the relationship the model shows.

My picture explanation, deciding evidence and one limit

04. Change one thing. Trace what follows.



Follow the labelled picture with the matching explanation. Open the vector diagram

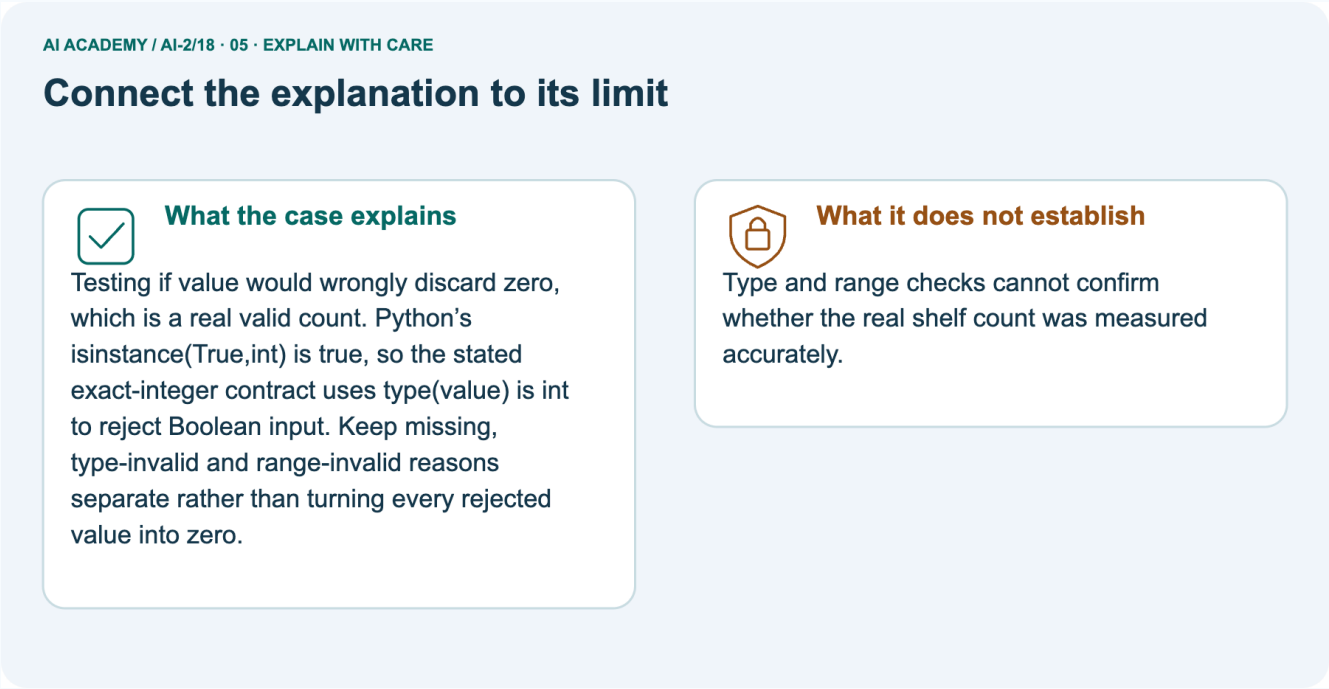
Original case: Accepted counts [0,4]; four rows rejected with recorded reasons. Changed condition: Replace text "4" with integer 4 through an authorised conversion.

Expected result: Accepted values become [0,4,4]; three rows remain rejected. Reason: The converted value now satisfies both type and range checks.

Predict before checking the worked changed-case result. Name what changed and what the supplied facts allow you to conclude.

My picture explanation, deciding evidence and one limit

05. Connect the explanation to its limit



Follow the labelled picture with the matching explanation. Open the vector diagram

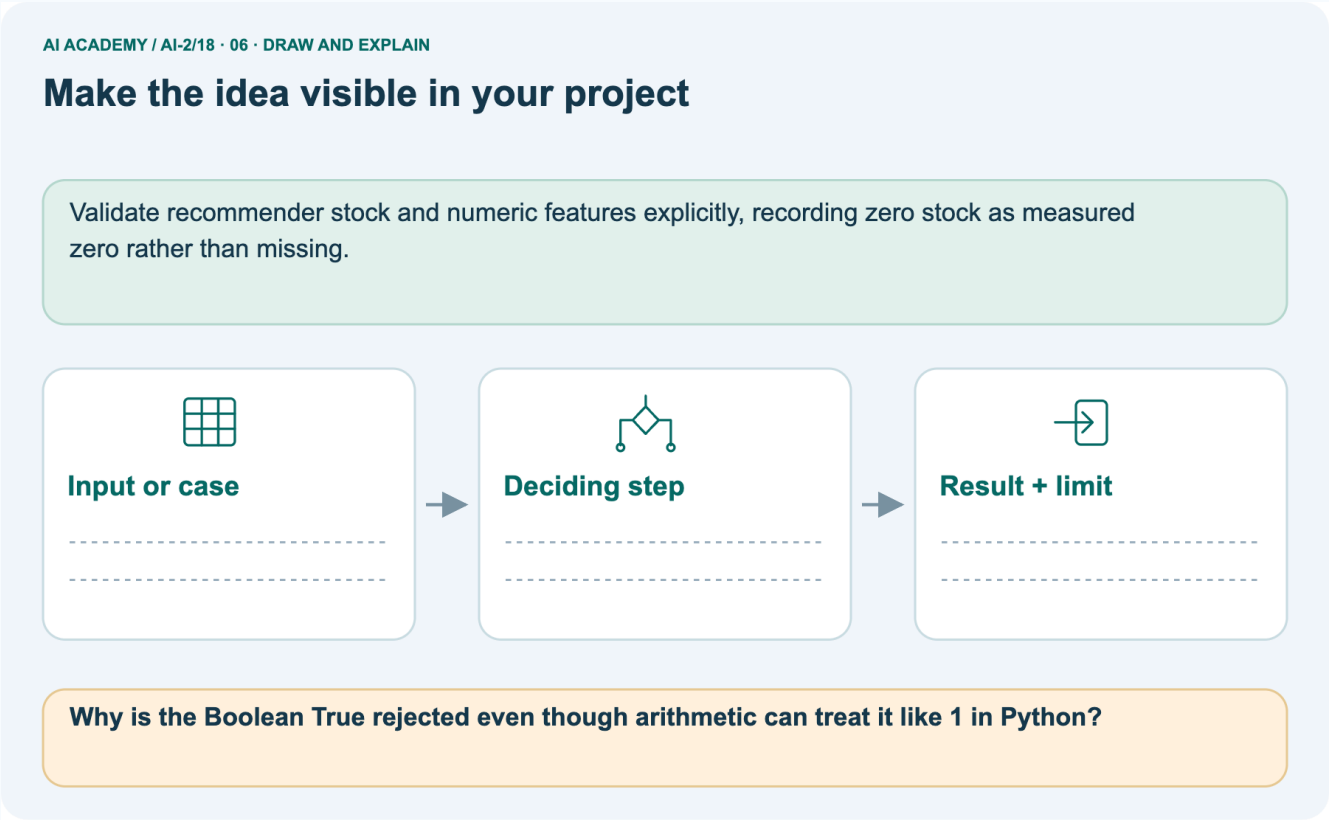
Testing if value would wrongly discard zero, which is a real valid count. Python's isinstance(True,int) is true, so the stated exact-integer contract uses type(value) is int to reject Boolean **input**.

Keep missing, type-invalid and range-invalid reasons separate rather than turning every rejected value into zero. Limit: Type and range checks cannot confirm whether the real shelf count was measured accurately.

Explain the case, then point to a limit on the claim.

My picture explanation, deciding evidence and one limit

06. Make the idea visible in your project



Follow the labelled picture with the matching explanation. Open the vector diagram

Validate recommender stock and numeric features explicitly, recording zero stock as measured zero rather than missing. Draw your **input**, deciding step, result and limit.

Supported question: Why is the Boolean True rejected even though arithmetic can treat it like 1 in Python?

Draw your project trace and justify the deciding step using **evidence**.

My picture explanation, deciding evidence and one limit

Save or print your responses for your portfolio. Progress and drafts on the hosted website stay in the current browser. Use only fictional or teacher-approved data.